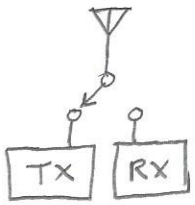
W2AEW

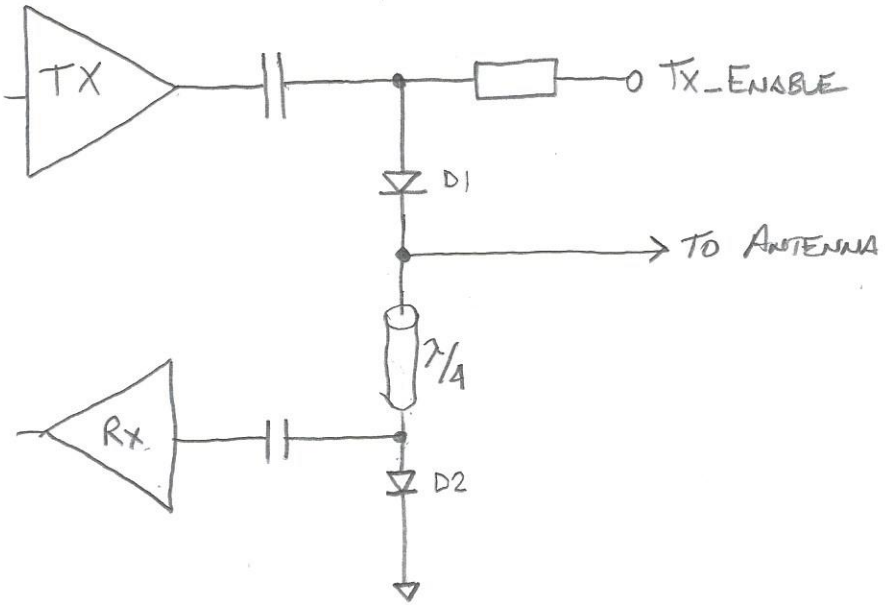
RF FUN

TRANSMIT / RECEIVE SWITCH (T/R SWITCH)



USING: - PIN DIODES

- QUARTER WAVELENGTH TRANSMISSION LINE



PIN DIODES

- LOW CAPACITANCE
- LONG REVERSE RECOVERY
- LOW FORWARD BIAS IMPEDANCE
- HIGH REVERSE BIAS IMPEDANCE

λ/4 TRANSMISSION LINE

- IMPEDANCE TRANSFORMER
- $\frac{Z_{IN}}{Z_0} = \frac{Z_0}{Z_L}$ $\quad Z_{IN} = \frac{Z_0^2}{Z_L}$
- LOOKS LIKE A SHORT WHEN OUTPUT IS OPEN
- LOOKS LIKE AN OPEN WHEN OUTPUT IS A SHORT

W2AEW

T/R SWITCH FOR 10.1 MHz (30m)

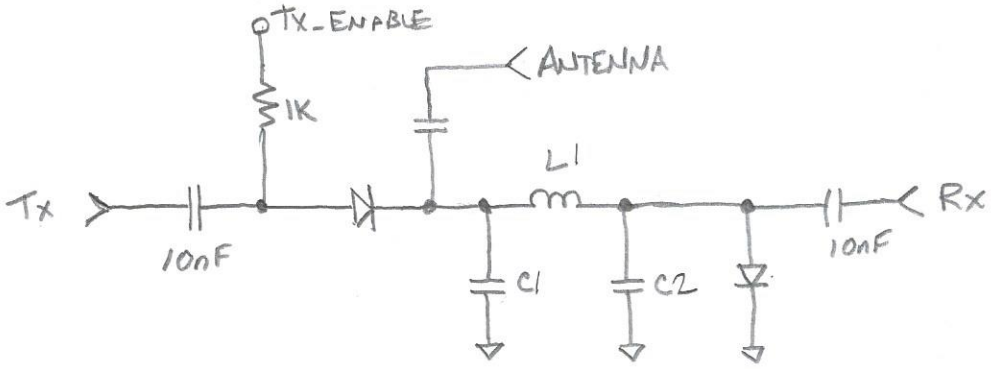
PROBLEM: $\lambda/4$ COAXIAL TRANSMISSION LINE

$\approx$ 16 FEET OR 4.9m LONG! (0.66 VF)

SOLUTION: USE A "LUMPED ELEMENT" EQUIVALENT

$X_L = X_C = Z_0$

use $Z_0 = 50$...

$$L = \frac{50}{2 \cdot \pi \cdot f} \approx 790nH$$

$$C = \frac{1}{2 \cdot \pi \cdot f \cdot 50} \approx 315pF$$